# Jihun Park

jihun999.github.io | Google Scholar | pjh2857@dgist.ac.kr | +82-10-4140-2857

## Contact

DGIST (Daegu Gyeongbuk Institute of Science and Technology)
Dept. Interdisciplinary Studies of Artificial Intelligence (ISAI)
E3-319, Techno jungang-daero 333, Hyeonpung-eup, Dalseong-gun, Daegu, Republic of Korea, 42988

## Summary

M.S.–Ph.D. integrated course student in Artificial Intelligence at DGIST, conducting research under the supervision of Prof. Sunghoon Im at the Robotics and Vision Lab (RVLab). My research centers on **representation alignment**—aligning features and attention across text, images, and reference subjects—so that generative models remain **consistent and controllable**. This perspective spans style transfer and object-centric editing, consistent multi-image generation, and reward-based post-training (ReFL, GRPO) for subject-driven personalization, and is currently being extended to **video world models**, where long-horizon temporal consistency and controllability are the central challenges. This line of work has been recognized with an **ACM SIGGRAPH Asia 2026 Oral**, an **AAAI 2026 Oral**, and a **CVPR 2025 Highlight (Top 3.7%)**, as well as multiple international peer-reviewed papers.

## Research Interests

- Representation Alignment for Generative Models
- Video World Models
- Image & Video Generation (Diffusion, Autoregressive)
- Style Transfer and Subject Personalization
- Vision–Language Understanding

## Education

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|---|---|
| Feb 2023 – Present | M.S. - Ph.D. Integrated Course, Interdisciplinary Studies of Artificial Intelligence (ISAI), **DGIST**, Daegu, South Korea. *Advisor: Prof. Sunghoon Im* |
| Sep 2018 – Jul 2022 | Bachelor of Mechanical Engineering, **Zhejiang University**, Hangzhou, China. |
| Mar 2015 – Feb 2018 | High School Diploma, **Chungnam Samsung Academy**, South Korea. |

## Work Experience

**Generative Model Research Intern, Baidu (Shenzhen, China)** *Dec 2025 – Mar 2026*

- Conducted research on image/video generation using large-scale diffusion and flow-matching models under the supervision of *Dr. Yan Zhang*.
- Explored reinforcement learning-based post-training (GRPO/ReFL) for generative model alignment, with reward function design combining perceptual and semantic metrics.
- Outcome: first-author paper (*CRAFT*) accepted to **ACM SIGGRAPH Asia 2026**. [paper] [project page]

## Selected Publications

* Equal contribution † Corresponding author (full list below)

- **Jihun Park**, Kyoungmin Lee, Jongmin Gim, Hyeonseo Jo, Jaeyeul Kim, Han Zou, Zhenpeng Zhan, Yan Zhang†, and Sunghoon Im†. "CRAFT: Constrained Reward via Attention Fine-Tuning for Subject Personal-

ization without Composed Targets", **ACM SIGGRAPH Asia**, Dec 2026. [Oral] [Collaborated with Baidu Global Business Unit, Shenzhen, China] [paper] [project page]

- Kyoungmin Lee*, **Jihun Park***, Jongmin Gim*, Wonhyeok Choi, Kyumin Hwang, Jaeyeul Kim and Sunghoon Im†. "A Training-Free Style-Personalization via SVD-Based Feature Decomposition", **CVPR**, Jun 2026. [paper]
- Minseok Oh*, **Jihun Park***, Jongmin Gim, Minwoo Choi, Kyoungmin Lee, Ferdinando Fioretto† and Sunghoon Im†. "FREESTYLE: An Anchor-Free Mechanism for Training-Free Style-Aligned Image Generation", **CVPR Findings**, Jun 2026. [paper]
- **Jihun Park***, Kyoungmin Lee*, Jongmin Gim*, Hyeonseo Jo, Minseok Oh, Wonhyeok Choi, Kyumin Hwang, Jaeyeul Kim, Minwoo Choi and Sunghoon Im†. "Infinite-Story: A Training-Free Consistent Text-to-Image Generation", **AAAI**, Jan 2026. [Oral] [Encouragement prize, 32nd HumanTech Paper Awards] [paper] [project page] [code]
- **Jihun Park***, Jongmin Gim*, Kyoungmin Lee*, Minseok Oh, Minwoo Choi, Jaeyeul Kim, Woochool Park, Yan Zhang, Zhenpeng Zhan, and Sunghoon Im†. "A Training-Free Style-aligned Image Generation with Scale-wise Autoregressive Model", *arXiv*, 2025. [Collaborated with Baidu Global Business Unit, Shenzhen, China] [paper]
- **Jihun Park***, Jongmin Gim*, Kyoungmin Lee*, Seunghun Lee, and Sunghoon Im†. "Style-Editor: Text-driven object-centric style editing", **CVPR**, Jun 2025. [Highlight paper, Top 3.7%] [Encouragement prize, 30th HumanTech Paper Awards] [paper] [project page] [code]

# Publications (Full List)

* Equal contribution † Corresponding author

**Peer-Reviewed**

- **Jihun Park**, Kyoungmin Lee, Jongmin Gim, Hyeonseo Jo, Jaeyeul Kim, Han Zou, Zhenpeng Zhan, Yan Zhang†, and Sunghoon Im†. "CRAFT: Constrained Reward via Attention Fine-Tuning for Subject Personalization without Composed Targets", **ACM SIGGRAPH Asia**, Dec 2026. [Oral] [Collaborated with Baidu Global Business Unit, Shenzhen, China] [paper] [project page]
- Sungho Moon*, Jaeyeul Kim*, Wonhyeok Choi, **Jihun Park**, Ukcheol Shin† and Sunghoon Im†. "Object-Centric LiDAR-to-Radar Distillation for 3D Object Detection", **IEEE Signal Processing Letters (SPL, IF: 3.9)**, Aug 2026. [paper]
- Kyumin Hwang*, Wonhyeok Choi*, Jaeyeul Kim, **Jihun Park**, Daehee Park†, and Sunghoon Im†. "CascadeOcc: Rethinking 3D Occupancy World Models with Cascaded VQ Representations", **IEEE Signal Processing Letters (SPL, IF: 3.9)**, May 2026. [paper]
- Jeonghoon Kim*, Hyeon Kang*, **Jihun Park**, Jinhwoi Kim, Jaeyeul Kim, and Sunghoon Im†. "Mitigating Noisy Correspondence in Video-Text Retrieval via Noise-mined Adaptive Self-Labeling", **ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM, IF: 6.1)**, May 2026. [paper]


- Kyoungmin Lee*, **Jihun Park***, Jongmin Gim*, Wonhyeok Choi, Kyumin Hwang, Jaeyeul Kim and Sunghoon Im†. "A Training-Free Style-Personalization via SVD-Based Feature Decomposition", **CVPR**, Jun 2026. [paper]


- Wonhyeok Choi, Kyumin Hwang, **Jihun Park**, Kyoungmin Lee, Seunghun Lee, Jaeyeul Kim, Minwoo Choi, and Sunghoon Im†. "TaskForce: Cooperative Multi-agent Reinforcement Learning for Multi-task Optimization", **CVPR**, Jun 2026. [paper]


- Minseok Oh*, **Jihun Park***, Jongmin Gim, Minwoo Choi, Kyoungmin Lee, Ferdinando Fioretto† and Sunghoon Im†. "FREESTYLE: An Anchor-Free Mechanism for Training-Free Style-Aligned Image Generation", **CVPR Findings**, Jun 2026. [paper]
- **Jihun Park***, Kyoungmin Lee*, Jongmin Gim*, Hyeonseo Jo, Minseok Oh, Wonhyeok Choi, Kyumin Hwang, Jaeyeul Kim, Minwoo Choi and Sunghoon Im†. "Infinite-Story: A Training-Free Consistent Text-to-Image Generation", **AAAI**, Jan 2026. [Oral] [Encouragement prize, 32nd HumanTech Paper Awards] [paper] [project page] [code]

- **Jihun Park***, Jongmin Gim*, Kyoungmin Lee*, Seunghun Lee, and Sunghoon Im†. "Style-Editor: Text-driven object-centric style editing", **CVPR**, Jun 2025. [Highlight paper, Top 3.7%] [Encouragement prize, 30th HumanTech Paper Awards] [paper] [project page] [code]
- Sanggyun Ma*, Wonjoon Choi*, **Jihun Park***, Jaeyeul Kim, Seunghun Lee, Jiwan Seo, and Sunghoon Im†. "Bridging Geometric and Semantic Foundation Models for Generalized Monocular Depth Estimation", **ICEIC**, Jan 2026. [paper]
- Sanggyun Ma*, Wonjoon Choi*, **Jihun Park**, Jaeyeul Kim, and Sunghoon Im†. "Semantic-Enhanced Monocular Depth Estimation via Fusion and Distillation of Foundation Models", **ICCVw**, Jul 2025. [paper]
- Jongmin Gim*, **Jihun Park***, Kyoungmin Lee*, and Sunghoon Im†. "Content-Adaptive Style Transfer: A Training-Free Approach with VQ Autoencoders", **ACCV**, Dec 2024. [paper]

**Preprints**

- **Jihun Park***, Jongmin Gim*, Kyoungmin Lee*, Minseok Oh, Minwoo Choi, Jaeyeul Kim, Woochool Park, Yan Zhang, Zhenpeng Zhan, and Sunghoon Im†. "A Training-Free Style-aligned Image Generation with Scale-wise Autoregressive Model", *arXiv*, 2025. [Collaborated with Baidu Global Business Unit, Shenzhen, China] [paper]

**Under Review**

- **Jihun Park**, Jongmin Gim, Kyoungmin Lee, Wonhyeok Choi, Minwoo Choi, Kyumin Hwang, Minseok Oh, and Sunghoon Im†. "ConsistKV: Long-Term KV Memory for Consistent Interactive World Models". **Under Review**, 2026.
- Tianyi Zheng, Minwoo Choi, **Jihun Park**, and Sunghoon Im†. "Understanding and Mitigating State Exposure Mismatch in Flow Model Reward Alignment". **Under Review**, 2026.
- Kyoungmin Lee*, Jongmin Gim*, **Jihun Park***, Minwoo Choi, Wonhyeok Choi, and Sunghoon Im†. "Style-Additive Editing: Cross-Domain Style Alignment for Image Editing Models". **Under Review**, 2026.
- Jongmin Gim*, Kyoungmin Lee*, **Jihun Park***, Wonhyeok Choi, Minwoo Choi, and Sunghoon Im†. "R2E-Prompt: Reasoning-to-Executable Prompt for Image Editing". **Under Review**, 2026.
- Minseok Oh, **Jihun Park**, Hyeonseo Jo, Jiwon Jang, and Sunghoon Im†. "Revise Once, View Everywhere: Text-Editable Single-Image Novel View Synthesis". **Under Review**, 2026.
- Minwoo Choi*, DongHyeon Kim*, Hyun SeungJun, Wonhyeok Choi, **Jihun Park**, and Sunghoon Im†. "MEF-FIT: Memory Efficient Trajectory Control for MM-DiT Based Video Diffusion Models". **Under Review**, 2026.
- Sungho Moon, Wonhyeok Choi, Seunghun Lee, **Jihun Park**, Jaeyeul Kim, Kyong Hwan Jin, and Sunghoon Im†. "ATTUNE: Adaptive Temporal and Text-Unified audio iNtEgration". **Under Review**, 2026.
- Minwoo Choi, Wonhyeok Choi, Jungwan Woo, Kyoungmin Lee, **Jihun Park**, and Sunghoon Im†. "Taming Generative Diffusion Models for Online Reinforcement Learning". **Under Review**, 2026.

## Awards

- Encouragement prize, 32nd HumanTech Paper Awards — Samsung Electronics Co., Ltd. *Feb 2026*
- Best Oral Presentation Award, 2025 DGIST Student Conference — DGIST, EECS/AI. *Oct 2025*
- Encouragement prize, 30th HumanTech Paper Awards — Samsung Electronics Co., Ltd. *Feb 2024*

## Teaching Experience

- Invited speaker of Introduction to Computer Vision (30+ attendees) — DGIST. *May 2026*
- Invited speaker of DGIST Generative AI Integrated Seminar (30+ attendees) — DGIST. *Oct 2024*
- Teaching Assistant (TA) of Advanced Deep Learning (80+ students) — DGIST. *Mar 2024 – Jun 2024*

## Academic Reviewer

- The Association for the Advancement of Artificial Intelligence (AAAI). *2025–present*
- International Conference on Learning Representations (ICLR). *2026–present*
- IEEE Transactions on Multimedia (TMM), *2026–present*

## Projects

**Multi prompt-based image generation (NIPA Innovation Hub AI Data Convergence Project)** *Jul 2024 – Present*

- Research about text-to-image diffusion models with fast sampling
- Improving the performance of image editing models

**Software development of smart glasses (Industry-Academic R&BD Collaboration)** *Jul 2023 – Jun 2024*

- Development of a vision-picking system for logistics based on AI object recognition
- Development of an object detection module and data processing

## Patents

- METHOD FOR GENERATING ANCHOR-FREE STYLE-ALIGNED IMAGES USING MAJORITY VOTING AND SET-BASED PROJECTION, COMPUTER PROGRAM STORED IN COMPUTER-READABLE RECORDING MEDIUM FOR EXECUTING THE SAME, AND APPARATUS FOR PERFORMING THE SAME (10-2026-0164003)
- METHOD FOR GENERATING PERSONALIZED IMAGE IN A NON LEARNING STYLE USING A SCALE-BASED AUTOREGRESSIVE MODEL. (10-2025-0099672)
- METHOD FOR DEPTH ESTIMATION BASED ON SEMANTIC INFORMATION THROUGH FUSION OF FOUNDATION MODELS AND KNOWLEDGE DISTILLATION. (10-2025-0099244)
- METHOD FOR GENERATING STYLE ALIGNED IMAGES USING AUTOREGRESSIVE MODEL. (10-2025-0054822)
- MONOCULAR DEPTH ESTIMATION METHOD BASED ON FUSION OF GEOMETRIC AND SEMANTIC INFORMATION. (10-2024-0176489)
- CONTENT-ADAPTIVE VECTOR QUANTIZATION-BASED NON-LEARNING STYLE SWITCHING TECHNIQUE, Publication date: Nov. 21, 2024. (10-2024-0166851)
- COMPUTER PROGRAM FOR TEXT-BASED, OBJECT-ORIENTED STYLE TRANSFER. (10-2023-0195850)
- COMPUTER PROGRAM AND METHOD FOR STYLE TRANSFER. Publication date: Mar. 02, 2025. (10-2023-0131272)
- COMPUTER PROGRAM AND METHOD FOR LOST AND FOUND SYSTEM. (10-2018-0072114)

## Other Experiences

- Exhibition of AI-Created Artworks at Kansong Art Museum, Daegu [Link] *Feb 2026 – May 2026*
- Exhibition of research on AI-driven art at DGIST [Link] *Nov 2025 – Feb 2026*
- Attended International Computer Vision Summer School (ICVSS 2025) in Sicily, Italy. *Jul 2025*
- Selected to represent DGIST at the official institutional booth during the 2025 Korea Science Festival. *Apr 2025*

## Skills

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| Languages | Korean (Native), Chinese (Fluent), English (Proficient) |
| Programming | Python, C, LaTeX |
| ML Frameworks | PyTorch, HuggingFace Diffusers, TensorFlow |
| Libraries | NumPy, Pandas, scikit-learn |